Computation of Bending Energy on The Boundary of a Triangulated Mesh

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June 2026

Motivation

Let us consider that we want to simulate a solid 2D membrane or ribbon. How do we study the elastic deformations of the membrane under thermal fluctuations? First, we figure out how to simulate it. Although the most accurate approach would be to try to compute the entire continuum structure, it is impossible for a computer to solve for a continuous physical field over an infinite number of points. How do we get around this problem? We discretise the structure into a mesh - specifically, a triangulated mesh. There are several reasons for this, firstly the fact that a triangle between any three given points in space is strictly planar and guaranteed to be unique. (This is also why computer graphics processing units specifically render solid shapes in the form of triangles.)

Say we discretise our membrane in the form of a triangulated mesh. Now, how do we study elastic deformations of said membrane? Specifically for our purpose, since we are using the MeMC [1] code as the baseline, we briefly outline how the MeMC code handles the initialisation of this membrane, in order to understand what we need to do.

1. Randomly initialise the coordinates of N points on a surface, here, we will be working with a flat surface, so we initialise the 2D coordinates constrained by a box according to what we want the dimensions of our membrane/ribbon to be.

2. Equilibriate the configuration with Monte Carlo simulations with a repulsive Lennard-Jones potential. Each displacement move is accepted only if it lowers the energy. Once the energy is minimized, we can be assured that the points are roughly evenly spaced along our box and we can triangulate it cleanly.

3. Finally, the triangulated mesh is generated by computing the Delaunay triangulation of the final configuration of points. The mesh data is stored in terms of the neighbour list, triangles, and position coordinates of the points.

Now, how do we simulate elastic deformations of this membrane?

Elastic Deformations of a 2D Membrane

We consider a 2D elastic membrane sitting in R^3. We parametrize the surface by using the Lagrangian (material) coordinates, i.e. labelling points on the material, which move with the material. The membrane can be described as a smooth map $\vec{R} : R^2 \to R^3$, mapping each point (x^1, x^2) to $\vec{R}(x^1, x^2)$.

The undeformed membrane is flat, i.e. it can be represented just by the two parametric coordinates, $\vec{R}(x^1, x^2) = x^1\hat{e_1} + x^2\hat{e_2}$.

Monge Parametrization: We will be working in the Monge gauge, or Monge parametrization, wherein we assume that the displacements due to the deformations are small enough that there is no overhanging or self-crossing, i.e. the surface remains single-valued. After deformation (x^1, x^2) moves to a new point in R^3. Within the Monge gauge, we denote the displacement as two in-plane comonents $u_\alpha(x^1, x^2)$ and one out-of-plane component $f(x^1, x^2)$, i.e.

$$\vec{R}'(x^1, x^2) = (x^\alpha + u_\alpha)\hat{e}_\alpha + f\hat{e_3}. \tag{1}$$

Strain Tensor [2]

To find the strain tensor, we want to track how the first fundamental form of the surface changes due to the deformation. Specifically, it is the change in the metric tensor we want in order to quantify the strain.
For a surface patch $\vec{R}(x^1, x^2)$, the metric tensor is given by $g_{\alpha\beta} = \partial_\alpha \vec{R} \cdot \partial_\beta \vec{R}$.
The first fundamental form on the undeformed surface,

$$\begin{aligned} ds^2 &= dx^1 dx^1 + dx^2 dx^2 + dx^3 dx^3 \\ &= dx^\alpha dx^\alpha = \delta_{\alpha\beta} dx^\alpha dx^\beta. \end{aligned} \tag{2}$$

The metric tensor on the undeformed surface is simply $\delta_{\alpha\beta}$.
On the deformed surface,

$$\partial_\alpha \vec{R} = (\delta_{\alpha\beta} + \partial_\alpha u_\beta)\hat{e_\beta} + \partial_\alpha f \hat{e_3}. \tag{3}$$

Therefore, the metric tensor,

$$\begin{aligned} g_{\alpha\beta} &= \partial_\alpha \vec{R} \cdot \partial_\beta \vec{R} \\ &= (\delta_{\alpha\gamma} + \partial_\alpha u_\gamma)(\delta_{\beta\gamma} + \partial_\beta u_\gamma) + \partial_\alpha f \partial_\beta f \\ &= \delta_{\alpha\beta}\delta_{\beta\gamma} + \delta_{\alpha\gamma}\partial_\beta u_\gamma + \delta_{\beta\gamma}\partial_\alpha u_\gamma + \partial_\alpha u_\gamma \partial_\beta u_\gamma + \partial_\alpha f \partial_\beta f \\ &= \delta_{\alpha\beta} + \partial_\alpha u_\beta + \partial_\beta u_\alpha + \partial_\alpha u_\gamma \partial_\beta u_\gamma + \partial_\alpha f \partial_\beta f. \end{aligned} \tag{4}$$

Therefore, the change in the metric tensor

$$g_{\alpha\beta} - \delta_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha + \partial_\alpha u_\gamma \partial_\beta u_\gamma + \partial_\alpha f \partial_\beta f. \tag{5}$$

We define $\frac{1}{2}(g_{\alpha\beta} - \delta_{\alpha\beta})$ as the strain tensor $s_{\alpha\beta}$. Now, we work in the limit such that in-plane displacements are small, as a consequence $\delta_\alpha u_\beta$ can be regarded as small enough that higher order terms of it can be neglected. Therefore, we drop the third term.
The strain tensor,

$$s_{\alpha\beta} = \frac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha + \partial_\alpha f \partial_\beta f). \tag{6}$$

The $\partial_\alpha u_\beta + \partial_\beta u_\alpha$ term is the linear in-plane strain. The last term shows us that we cannot have pure bending without stretching - having a pure out-of-plane displacement still introduces an in-plane component of the strain.

Curvature Tensor [3]

The curvature tensor is given by the second fundamental form of the surface, which tells us how fast our tangent directions are changing along the surface, i.e. how much the surface is curving.
Let us write the basis vectors $\partial_\alpha \vec{R}$ of the tangent plane as $\vec{a}_\alpha$. We can correspondingly define the dual basis $\vec{a}^\alpha$, such that $\vec{a}^\alpha \cdot \vec{a}_\beta = \delta^\alpha_\beta$.
The curvature tensor is defined by the outer product,

$$K = -\frac{\partial \vec{n}}{\partial x^\alpha} \otimes \vec{a}^\alpha, \tag{7}$$

where $\vec{n}$ is the unit normal vector to the surface.
(Differentiating $\vec{n} \cdot \vec{n} = 1$ with respect to x^α, we get $\vec{n} \cdot \frac{\partial \vec{n}}{\partial x^\alpha} = 0$, i.e. $\frac{\partial \vec{n}}{\partial x^\alpha}$ lies in the tangent plane.)
Now,

$$\begin{aligned} \frac{\partial \vec{n}}{\partial x^\alpha} &= -K_{\beta\alpha}\vec{a}^\beta \\ \implies \vec{a}_\gamma \cdot \frac{\partial \vec{n}}{\partial x^\alpha} &= -K_{\beta\alpha}\vec{a}_\gamma \cdot \vec{a}^\beta \\ \implies \vec{a}_\gamma \cdot \frac{\partial \vec{n}}{\partial x^\alpha} &= -K_{\alpha\beta}\delta^\beta_\gamma \\ \implies \vec{a}_\gamma \cdot \frac{\partial \vec{n}}{\partial x^\alpha} &= -K_{\alpha\gamma}. \end{aligned} \tag{8}$$

Using the last result to find the elements of K,

$$\begin{aligned}K_{\alpha\beta} &= -\frac{\partial\vec{n}}{\partial x^\alpha}\cdot\vec{a}_\beta \\ &= -\frac{\partial}{\partial x^\alpha}(\vec{n}\cdot\vec{a}_\beta)+\vec{n}\cdot\frac{\partial\vec{a}_\beta}{\partial x^\alpha} \\ &= \vec{n}\cdot\frac{\partial\vec{a}_\beta}{\partial x^\alpha} \\ &= \vec{n}\cdot\frac{\partial^2\vec{R}}{\partial x^\alpha x^\beta}.\end{aligned} \tag{9}$$

Therefore, the curvature tensor is given by,

$$K_{\alpha\beta} = \vec{n}\cdot\partial_\alpha\partial_\beta\vec{R}. \tag{10}$$

Now, the normal vector is perpendicular to both tangents, i.e.

$$\vec{n} = \frac{\partial_1\vec{R}\times\partial_2\vec{R}}{\left|\partial_1\vec{R}\times\partial_2\vec{R}\right|} \tag{11}$$

In the Monge gauge, neglecting second order derivatives of the in-plane displacements, we have $\partial_\alpha\vec{R} = \hat{e}_\alpha + \partial_\alpha f\hat{e}_3$.
The cross product,

$$\partial_1\vec{R}\times\partial_2\vec{R} = \hat{e_1}(-\partial_1 f)+\hat{e_2}(-\partial_2 f)+\hat{e_3}, \tag{12}$$

and $|\partial_1\vec{R}\times\partial_2\vec{R}| = \sqrt{1+(\partial_1 f)^2+(\partial_2 f)^2}$.
Now,

$$\partial_\alpha\partial_\beta\vec{R} = (\partial_\alpha\partial_\beta u_\gamma)\hat{e_\gamma}+(\partial_\alpha\partial_\beta f)\hat{e_3}\approx\partial_\alpha\partial_\beta f\hat{e_3}, \tag{13}$$

where we neglect higher-order derivatives of the in-plane displacements.
So the curvature tensor is given by the inner product with $\vec{n}$,

$$K_{\alpha\beta} = \frac{\partial_\alpha\partial_\beta f}{\sqrt{1+(\partial_1 f)^2+(\partial_2 f)^2}}. \tag{14}$$

We keep terms upto second order in f, considering that $\partial_\alpha f$ is small enough. Then the curvature tensor simplifies to,

$$K_{\alpha\beta} = \partial_\alpha\partial_\beta f. \tag{15}$$

Gaussian and Mean Curvatures [2]

We consider the principal direction, i.e. along which K is diagonal (we know K is diagonalizable because it is real and symmetric). The eigenvalues κ_1 and κ_2 are the principal curvatures, and are the maxima and minima of the normal curvature taken along all possible directions.
The mean curvature is defined as,

$$H = \frac{1}{2}(\kappa_1+\kappa_2), \tag{16}$$

and the Gaussian curvature is defined as,

$$K_G = \kappa_1\cdot\kappa_2. \tag{17}$$

We note that the mean and Gaussian curvatures are half the trace, and the determinant, of the curvature tensor matrix, respectively.
In the Monge gauge,

$$H = \frac{1}{2}(\partial_1^2 f+\partial_2^2 f) = \frac{1}{2}\nabla^2 f, \tag{18}$$

and,

$$K_G = \partial_1^2 f\partial_2^2 f-(\partial_1\partial_2 f)^2. \tag{19}$$

Bending Energy

(For now, I am writing the argument, instead of a rigorous derivation, for the bending energy.)
We note, translation or rotation of the surface would not give rise to any bending. So the bending energy should not depend on the first or second derivative of the surface equation.
When we introduce a curvature to the surface, bending occurs. Therefore, the bending energy shoul depend purely on $\partial_\alpha \partial_\beta \vec{R}$ i.e. the curvature tensor.
Since the energy is a scalar, it would be invariant to any rotational transformation of the coordinate axes. Therefore, it must depend on properties of K that are also invariants of such a transformation. These properties are the trace and the determinant.
Further, we note that, for a homogeneous sheet (such that both sides are of the same material), the bending energy should not be dependent on the direction of bending. So, the sign of the mean curvature should not matter, i.e. the energy should be a function of the $Tr(K)^2$, not just $Tr(K)$. Therefore, the general bending energy density would look like,

$$F = a(2H)^2 + bK_G. \tag{20}$$

Integrating the energy over the volume of the sheet and using the expressions for the actual elastic moduli, we end up with an integral over the area after the height is integrated out. Using the fact that $2H = \nabla^2 f$ in the Monge gauge, the total bending energy is given by,

$$H_{bend} = \int dA \left(\frac{\kappa}{2} (\nabla^2 f)^2 + \bar{\kappa} K_G \right), \tag{21}$$

where κ is the bending modulus and $\bar{\kappa}$ is the saddle-splay modulus.

Computing the Energy on Our Mesh

We have two terms in the energy calculation which we need to implement on our triangulated mesh. For the discrete mesh, the properties of the surface at each vertex are defined as spatial averages around the vertex, on the 1-ring neighbourhood of the vertex.
The first term is the Laplacian of the out-of-plane displacement. This is already implemented in MeMC. The discrete form of the integral of the Laplacian is given by [4]

$$\int_A L_i dA_i = \frac{1}{2} \sum_{j \in N_1(i)} (\cot \alpha_{ij} + \cot \beta_{ij})(\vec{x_i} - \vec{x_j}), \tag{22}$$

where α_{ij} and β_{ij} are the two angles opposite to the edge in the two triangles sharing the edge (x_i, x_j), and $N_1(i)$ is the total 1-ring neighbourhood surrounding the vertex i.

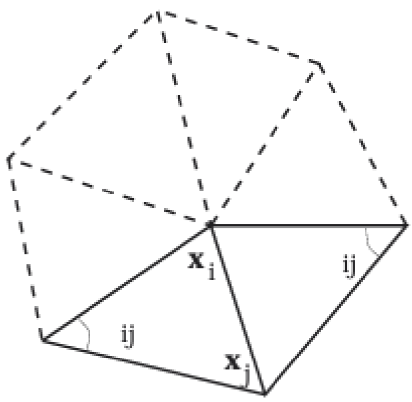

Figure 1: Notation of the angles and edges for calculation of the Laplacian and the Voronoi areas.

The area of the Voronoi dual cell is computed according to whether the triangle where the Voronoi region falls is acute or obtuse. For triangles having no obtuse angle, we use the Voronoi area,

$$A_i^{\text{Voronoi}} = \frac{1}{8} \sum_{j \in N_1(i)} (\cot \alpha_{ij} + \cot \beta_{ij}) \left\| \mathbf{x}_i - \mathbf{x}_j \right\|^2 \tag{23}$$

. For obtuse triangles, the Voronoi region may or may not lie inside the triangle. We use, for an obtuse triangle T, if the angle at the vertex is obtuse, $area(T)/2$, and if any other angle is obtuse, $area(T)/4$, to add in the total area A.
Thus, the discrete Laplacian (and twice the mean curvature H) is computed directly as,

$$L_i = \frac{1}{2A} \sum_{j \in N_1(i)} (\cot \alpha_{ij} + \cot \beta_{ij})(\vec{x_i} - \vec{x_j}). \tag{24}$$

The second term is the integral of the Gaussian curvature over the area. This is what we need to implement. The existing MeMC algorithm has provision for either closed boundary, or clamped boundary conditions. Closed boundary condition implies there are no boundary vertices, e.g. in a sphere, and clamped boundary means we simply exclude the boundary vertices from the Monte Carlo updates, as a result of which we do not need to handle the energy at the boundary vertices at all. Why does the Gaussian curvature term disappear from our computation in these cases?

Gauss-Bonnet Theorem

The Gauss-Bonnet Theorem states that, for a manifold M with a boundary which we denote by m,

$$\int_M K_G dA + \int_m \kappa_g ds = 2\pi\chi(M), \tag{25}$$

where $\chi(M)$ is the Euler characteristic of the manifold, and κ_g is the geodesic curvature.
The Euler characteristic depends only on the topology of the surface, and for elastic deformations, does not change. For a surface that does not have an open boundary, the geodesic curvature term simply vanishes. As a result the Gaussian curvature term is an irrelevant constant, which does not affect the energy change in our Monte Carlo updates.
Now, when we are trying to implement a stress-free boundary, we can no longer disregard the Gaussian curvature term.

Strategy: To compute the second term, i.e. area integral of the Gaussian curvature, from the line integral of the geodesic curvature over the boundary.

So, the problem of how to handle the bending energy for a stress-free boundary factors into two questions:

1. What happens to the Laplacian term at the boundary, where the 1-ring is not complete?
2. How to calculate the geodesic curvature at the boundary?

Geodesic Curvature

The geodesic curvature of a curve γ in a manifold $\bar{M}$ constrained to lie on a submanifold M, is defined as the projection of the curvature vector $D\vec{T}/ds$, (where T is the unit tangent vector and s is the arclength) onto the tangent space of the submanifold,

$$\kappa_g = \frac{d\vec{T}}{ds}(\vec{N} \times \vec{T}), \tag{26}$$

where $\vec{N}$ is the unit normal of the surface, and $\vec{T}$ is the unit tangent.
The geodesic curvature essentially tells us how far the curve is from being a geodesic on the surface. The curvature of a curve has two components: one due to the bending of the surface itself. This component is given by the normal curvature. The other component is the curvature of the curve within the surface, this is the geodesic curvature.

Examples:

1. A Circle on a Cylinder
1.1. We first consider a cirle as a simple space curve. The curvature of a space curve is given by

$$\kappa = \left|\left|\frac{d\vec{T}}{ds}\right|\right|, \tag{27}$$

where $\vec{T}$ is the unit tangent vector of the curve and s is the arc-length.
The parametric equation of a circle is,

$$\vec{\gamma}(t) = (R\cos t, R\sin t, 0). \tag{28}$$

The tangent vector,

$$\frac{d\vec{\gamma}}{ds} = \frac{1}{R}(-R\sin t, R\cos t, 0) = (-\sin t, \cos t, 0), \tag{29}$$

using $ds = Rdt$. This is already a unit vector.
Again,

$$\frac{d\vec{T}}{ds} = \frac{1}{R}(-\cos t, -\sin t, 0). \tag{30}$$

$$\kappa = \left|\left|\frac{d\vec{T}}{ds}\right|\right| = \frac{1}{R}\sqrt{\cos^2 t + \sin^2 t} = \frac{1}{R}. \tag{31}$$

1.2. Now, we compute the geodesic curvature of the same circle as the open edge of a cylinder. We have, from the previous part,

$$\frac{d\vec{T}}{ds} = \frac{1}{R}(-\cos t, -\sin t, 0). \tag{32}$$

Now, the surface normal for a vertical cylinder,

$$\vec{N} = (\cos t, \sin t, 0). \tag{33}$$

The cross product $\vec{N} \times \vec{T} = (0, 0, \cos^2 t + \sin^2 t) = (0, 0, 1)$. Then, $(d\vec{T}/ds) \cdot (\vec{N} \times \vec{T}) = 0$, i.e. the geodesic curvature is 0.
So, the entire curvature of the circle is in the normal direction. On the surface of the cylinder, the edge circle is a geodesic.

Computing the Geodesic Curvature on the Boundary

We are interested in computing the geodesic curvature specifically over the boundary vertices.

1. Discrete Geodesic Curvature The discrete geodesic curvature of a curve on a polyhedral mesh is defined to be [5, 6]

$$\kappa_g = \frac{2\pi}{\theta}\left(\frac{\theta}{2} - \theta_r\right), \tag{34}$$

where θ_l and θ_r are the total right and left angles made by the curve at the edge it intersects.

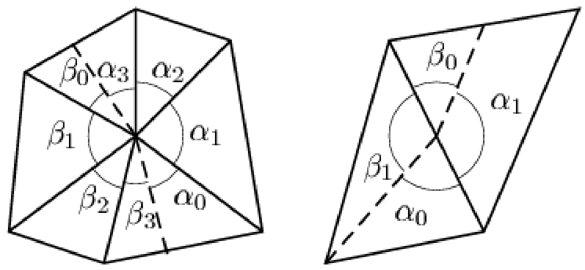

Figure 1. Right and left angles (θ_r and θ_l resp.) in a curve. $\theta_r = \sum \alpha_i$ and $\theta_l = \sum \beta_i$.

Figure 2: Computation of discrete geodesic curvature of an interior curve on a triangulated mesh.

The Issue: This is defined for a curve on the interior of the mesh. Our purpose is to calculate κ_g on the boundary itself.
The Idea: We consider a curve segment on the boundary, formed by 3 vertices ABC. We want to calculate the geodesic curvature at B. We shift a copy of the segment ABC by a distance ϵ inwards, now this $A'B'C'$ is an interior curve which intersects the edges. We now compute the discrete geodesic curvature of $A'B'C'$, and then iterate for smaller and smaller ϵ, and extrapolate to find the limiting value of the discrete geodesic curvature as ϵ approaches 0.

2. Using the Definition of Eq. 26
If we can determine the tangent and normal at the surface, we can directly use the formula for geodesic curvature. The tangent vector of the boundary curve at a vertex can be determined from the boundary edges of the triangles, leaving and entering the vertex.
The Issue: The normal of a curve is not unique in itself. Here, we compute the curve as a part of the surface, and hence we consider the surface normal. How is the normal defined at a vertex?
We use the definition of the surface normal at a vertex as the weighted average of the normal vectors of its adjacent triangles, weighted by their area, [7]

$$\vec{n}(i) = \frac{\vec{N}_i}{|\vec{N}_i|}, \quad \vec{N}_i = \sum_{j(i)} \vec{n}_{j(i)} A_{j(i)}, \tag{35}$$

where $j(i)$ represent the triangles connected to the vertex i.
Then, using the normal and the tangent expressions, we try to compute the geodesic curvature directly from Eq. 26.
(Is taking just the nearest adjacent points sufficient to effectively estimate both $\vec{T}$ and $d\vec{T}/ds$?)

Tasks

We want to test how well each of these approaches work on some test cases where the analytical result is known. Specifically, we will test on

1. The open edge of a cylinder (κ_g calculated to be 0).
2. The edge of a Mobius strip.

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